

Smart Home AIoT – Test Data Collection Sheet – Phase D – Safety Auto-Flows Simulation

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Instructions for use:

1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:

- Pass Rate = $(\text{Pass Count} / 10) \times 100\%$
- Avg RT = $(\text{response times}) / 10$
- Variance = $(x_i -)^2 / n$
- Std Dev = $\sqrt{\text{variance}}$

Coverage Matrix

Phase	Description	Count
Phase D	Phase D – Safety Auto-Flows Simulation	12
	TOTAL	12

Phase D – Safety Auto-Flows Simulation

D01	Gas Alert – Stove OFF				Phase D		
Method: mosquitto_pub home/kitchen/gas_alert ON							
Expected: switch.kitchen_stove OFF (double-block)							
#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10		Pass Rate: ____%		Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

D02

Gas Alert – Vent Fan ON

Phase D

Method: gas_alert trigger

Expected: switch.kitchen_fan ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: gas_alert trigger

Expected: script.living_room_red_flash fired (Tuya แดงพริบ)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

D04

Gas Alert – LINE Message

Phase D

Method: gas_alert trigger

Expected: LINE message " ก๊าซรั่ว!" arrives

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: แตะน้ำที่ probe / force binary_sensor.bathroom_water_leak = on

Expected: blue pulse + LINE " น้ำรั่ว!"

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: force binary_sensor.bathroom_fall_alert = on

Expected: scene.livingroom_pre_fall_alert saved

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: fall_alert trigger

Expected: ไฟแดงพริบ 30 วินาที (10 รอบ)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: fall_alert trigger

Expected: LINE " คนล้ม!" + scene restore กลับ

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

D09

Garage Smoke Alert

Phase D

Method: mosquitto_pub home/garage/smoke_alert ON

Expected: Red flash + LINE " คำนโรรงรณ!"

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: Set Security + force door open

Expected: Auto-lock + Red flash + LINE " ฝิดนญก!"

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: mosquitto_pub home/garden/soil_moisture 25

Expected: switch.garden_water_pump ON + LINE " ดินแห้ง"

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: stove ON 10/15 min + Away

Expected: LINE warning + auto-OFF

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments: